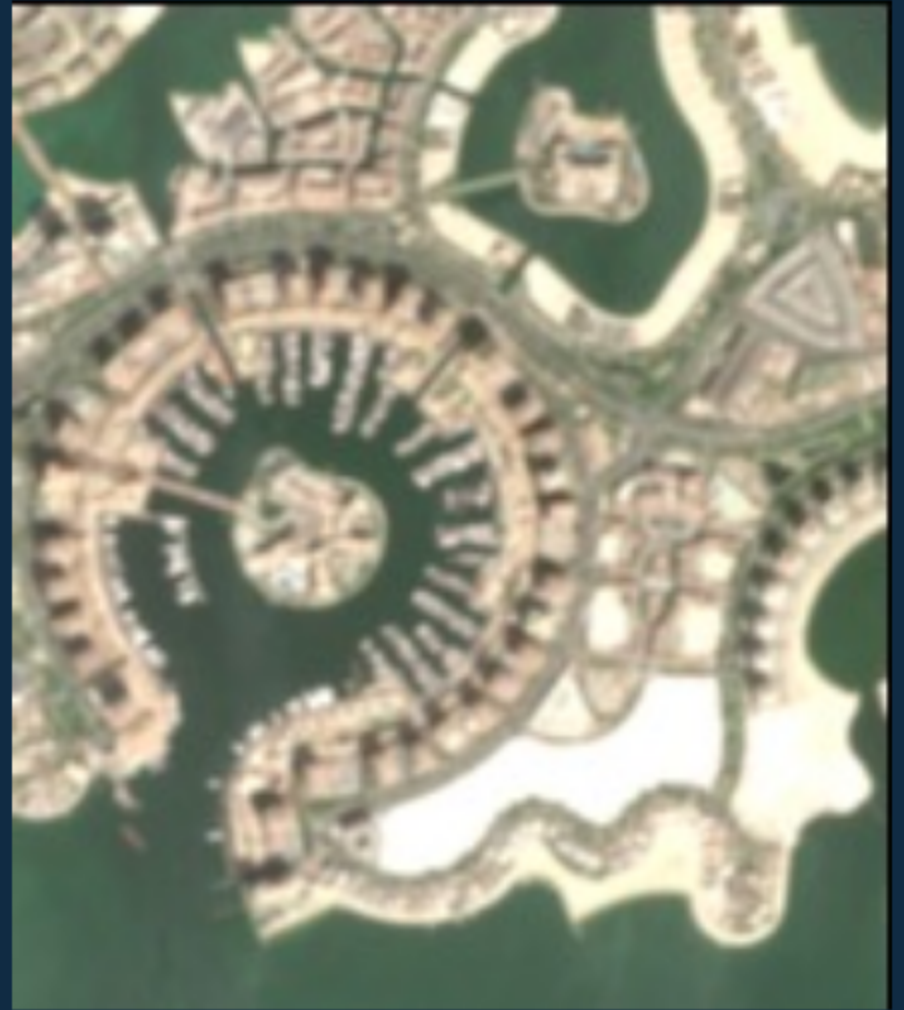


UrbanHeat AI

The Pearl planning pilot

Al Zubarah · Qatar

Urban Expansion, Land Use Change & Heat Risk



Open-data PoC · October 2026

The planning problem

Urban planners need a spatial starting point for heat-mitigation site visits.

Our pilot screens a small part of The Pearl using repeatable satellite observations and mapped urban features.

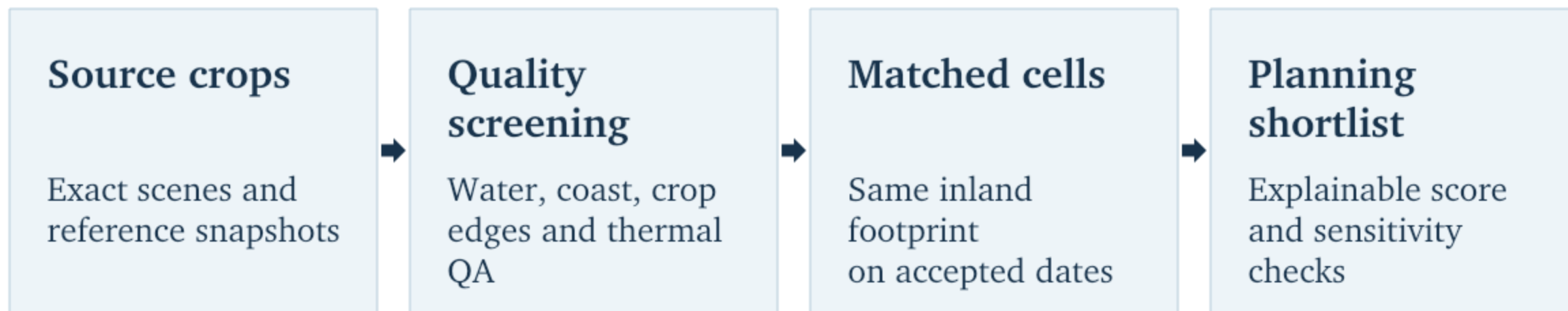
The decision: where should a planner inspect pedestrian use, existing shade and planting feasibility first?

Data and implemented scope

Source	Date	Used for
Sentinel-2 L2A	5, 15, 30 Sep 2026	Green signal and optical quality
Landsat 8/9 L2	6, 14, 30 Sep 2026	Surface temperature and QA
ESA WorldCover v200	2021	Historical ML-derived land cover
OpenStreetMap	Downloaded 4 Oct 2026	Mapped buildings and references

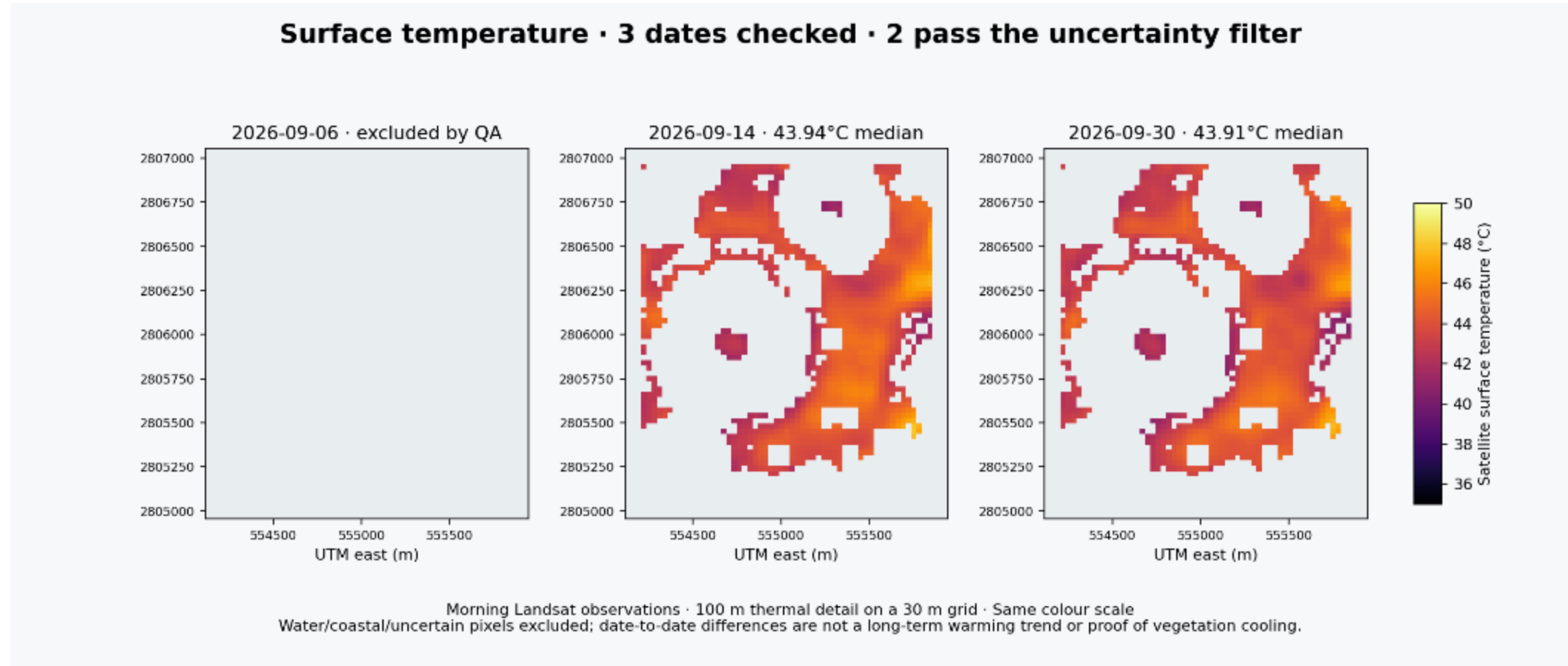
Open data. No VHR, 813 or hyperspectral imagery. We use a precomputed ML product and contributed geometry.

Analysis workflow



Thermal detail: about 100 m. Reporting cells: 300 m. Retained delivered pixels: 30 m and spatially dependent.

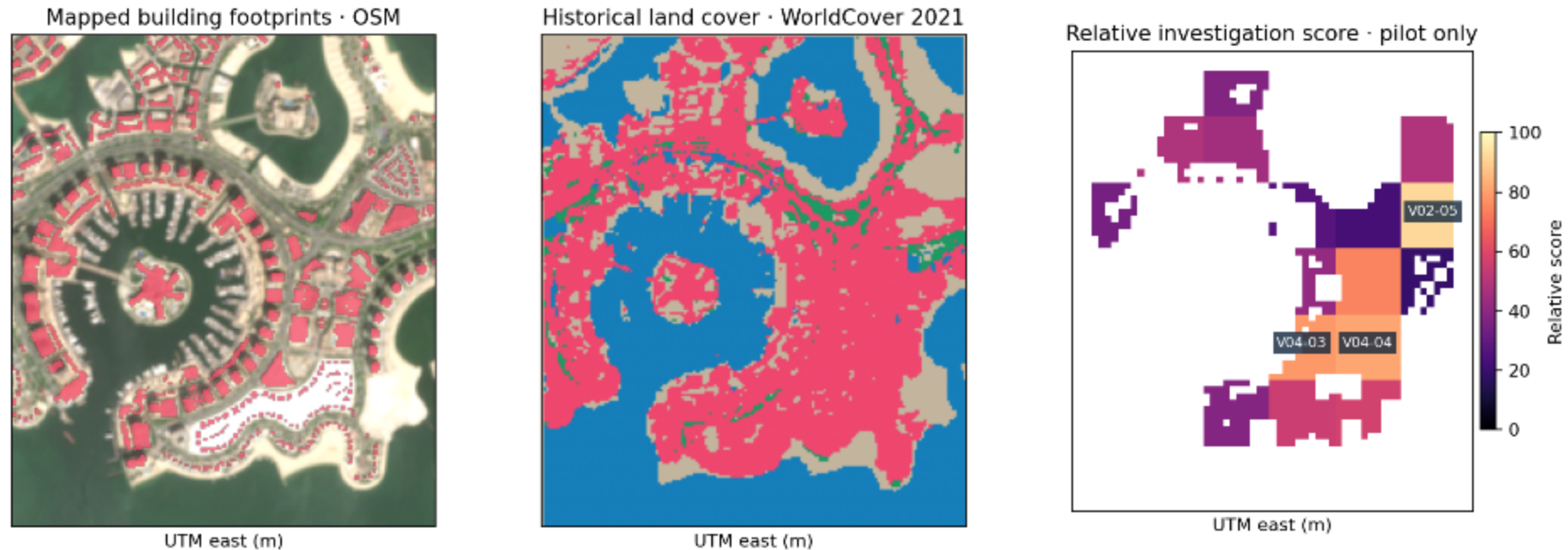
Accepted surface-temperature observations



1.15 km² retained on the same footprints. Surface temperature is distinct from air temperature and human heat exposure.

Land cover and investigation priorities

UrbanHeat AI · land-cover baseline and planning shortlist



WorldCover: blue water, pink built-up, green vegetation, beige other · OSM footprints are contributed and may be incomplete
Shortlist requires site checks for pedestrian use, shade, ownership and planting feasibility. No health-risk or cooling benefit inferred.

Historical land cover and contributed footprints provide context for site inspection.

Shortlist and sensitivity

Cell	Surface °C*	Green %*	Buildings %	Rank range
V02-05	45.2	5.2	34.9	1–1
V04-04	44.9	6.0	24.7	2–3
V04-03	44.2	4.2	30.3	2–6

**The first two candidates remain near the top across all 11 tested settings.
The third is less stable.**

*Averages of the two date-level cell summaries. Coverage is partial and reported per cell. No cooling benefit is estimated.

Reference checks and remaining limits

30 / 30 coastal-water points agree with the water screen

Small water features: all 3 sampled interiors are missed.

Grass: no interiors survive the reference erosion.

Buildings: 1 of 30 sampled interiors passes the green threshold.

Human reference confirmation and independent thermal calibration remain pending. These checks cannot certify overall accuracy.

Impact and decision limits

The output helps organise targeted site visits

A planner can inspect the shortlist, its contributing indicators, coverage and sensitivity before choosing local work.

No measured savings, intervention benefit, heat-health risk, rural-reference UHI intensity or long-term warming trend is claimed.

Next steps and submission readiness

Next scientific work

Confirm reference samples with a human reviewer. Add suitable VHR segmentation and paving/shade indicators when the data support them. Validate thermal observations independently.

Before submitting

Run the notebook in normal Jupyter on another machine. Confirm member names and roles, GitHub access and the current deadline. Submit the form with this PDF after team review.

Sources and licences

Contains modified Copernicus Sentinel data (2026)
Copernicus Sentinel terms. Exact scene records are in the repository.

Landsat imagery courtesy of the U.S. Geological Survey
Landsat 8–9 Collection 2 L2. doi.org/10.5066/P9OGBGM6

ESA WorldCover 2021 v200, Zanaga et al. (2022)
CC BY 4.0. doi.org/10.5281/zenodo.7254220

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